

Composite Disturbance Rejection Attitude Control for Quadrotor With Unknown Disturbance

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Abstract—In this article, a composite disturbance rejection control method is proposed for attitude control of quadrotor using an inner–outer loop control framework, which can effectively improve the angular tracking performance of the quadrotor and achieve the active disturbance rejection. In the inner loop, a novel nonlinear extended state observer (NESO) is designed to estimate the disturbance and nonlinear part of the system model, then a robust control law is embedded in inner-loop controller to attenuate the effect of the estimation error. In the outer loop, a composite nonlinear feedback (CNF) controller is designed to improve the transient performance. Moreover, the stability of two loops are also analyzed, respectively. Numerical simulation and platform experiment are carried out to verify the efficiency of the proposed method.

Index Terms—Attitude control, composite nonlinear feedback (CNF) controller, nonlinear extended state observer (NESO), quadrotor.

I. INTRODUCTION

IN RECENT years, unmanned aerial vehicles (UAVs) have received great attention due to their low cost, small size, and easy to repair. As one of the representations among UAVs, quadrotor is widely used in field of agriculture, industry, architecture, military, mapping, navigation, transportation, entertainment, and so on [1]–[3]. The quadrotor is a typical unstable under-actuated system, and the model is nonlinear with highly coupled states, which leads to the control of quadrotor a challenging problem [4], [5].

The attitude control is the primary and basic concern in quadrotor applications, and amounts of control approaches

have been proposed in recent years, for example, proportional-integral-derivative (PID) control [3], [4], [6], backstepping control [7], robust and adaptive control [8]–[10], and the references therein. It should be noted that, in the real flight applications, quadrotors are always affected inevitably by disturbances and uncertainties, such as gust wind, unmodeled dynamics, parametric uncertainties, and time delays, which may destroy the control performance. Therefore, it is important to consider the disturbance attenuation and active rejection in the quadrotor attitude control design to achieve better control performance. To address those issues, some disturbance attenuation control strategies have been proposed. A fuzzy-Pade controller for quadrotor attitude control is investigated in [11], which is a simplification and improvement of fuzzy logic controller. In [12], a nonlinear PID-type controller is proposed for quadrotor trajectory tracking, the hyperbolic tangent function is used to achieve a better performance in control accuracy and disturbance attenuation. In [13], by dividing the quadrotor model into a fully actuated and an underactuated subsystems, the sliding mode controllers are established for each subsystems. By combining the sliding mode control (SMC) and backstepping methods, a robust nonlinear controller is proposed for attitude and position trajectory tracking of quadrotor in [14]. In [15], with the help of finite-time sliding mode observer, an output feedback robust tracking control approach is proposed for a quadrotor with external disturbances. The effect of parametric uncertainties are considered in [16], and an adaptive sliding mode controller is designed for quadrotor stabilization in finite time. In [17], a switching model predictive controller (MPC) is designed for the trajectory tracking problem under the presence of wind gusts and system constraints. By utilizing an implicit model to generate output predictions and compute control action, an adaptive MPC is proposed for quadrotor altitude control in [18]. In [19] and [20], a robust attitude stabilization controller including a state feedback term and a robust compensator is designed, it takes quadrotor internal uncertainties and input delays into consideration. Besides, in [21], the continuous multivariable attitude control approach is proposed for quadrotor UAV, and the attitude tracking errors of quadrotor UAV can be driven to zero in finite time.

Different from the aforementioned disturbance attenuation control approaches, the active disturbance rejection control (ADRC) attracts great interest in control theory and in practice research areas, because of its strong robustness and less

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relying on precise system model. The key idea of ADRC is to estimate the disturbance by using extended state observer (ESO), and incorporate the disturbance estimates in the controller to actively compensate for the disturbance. In [22], the structure and each component of ADRC are presented in details. More recently, ADRC approaches have been developed for attitude control of quadrotor subject to disturbances. In [23], a fault tolerant controller is designed for dealing with quadrotor attitude stabilization problem. The total disturbance is composed by actuator faults, wind gusts, and measurement noises, which would be estimated and compensated by an improved ESO. In [24], a dual closed-loop attitude control framework is presented, and the ADRC controller is used in inner loop to compensate for the unknown dynamic of the quadrotor system and gust wind. Another dual closed-loop integrated ADRC control scheme is proposed in [25], aiming on the quadrotor trajectory tracking control system with nonlinearity, strong coupling, and external disturbance. It is worthy of mentioning that, the disturbance estimation accuracy of ESO plays an important role in ADRC design. In [23] and [24], the traditional nonlinear ESO (NESO) proposed in [22] is used for attitude control design of quadrotors, and there are number of parameters in the NESO to be tuned. In [25], instead of NESO, a linear ESO (LESO) is used in the attitude control of quadrotors and the tuning parameters of ESO are reduced. It should be pointed out that, although the LESO is superior to the NESO in parameter tuning and theoretical analysis, the LESO may be less effective than the NESO. How to improve the effectiveness of LESO and the attitude control performance of quadrotors is still an interesting work.

Motivated by these preceding results, in this article, the composite disturbance rejection control strategy is proposed for attitude control of quadrotor subject to unknown disturbances in an inner–outer loop control structure. More specifically, to improve the disturbance estimation performance, a novel NESO is designed in the inner loop, the convergence of the proposed ESO is analyzed, and a robust law is incorporated in the inner-loop controller to compensate the estimation error generated by ESO, which guarantees the states of the inner-loop control system are uniformly ultimately bounded (UUB). Moreover, in the outer loop, a CNF controller is introduced to improve the tracking speed of attitude system. The main contributions are summarized as follows.

- 1) By incorporating a nonlinear function in the LESO, a novel NESO is designed to improve the disturbance estimation performance, and the rigorous convergence analysis of proposed NESO is performed.
- 2) Under the inner–outer loop control structure, by incorporating the disturbance estimate in the inner-loop controller and applying a CNF controller in the outer loop, the composite disturbance rejection control approach is proposed to improve the attitude control precision.

Finally, simulation and experimental results are presented to show the effectiveness of the proposed attitude control approach.

II. SYSTEM MODELING

In this section, the kinematic and dynamic models of quadrotor are introduced. The quadrotor is an underactuated system

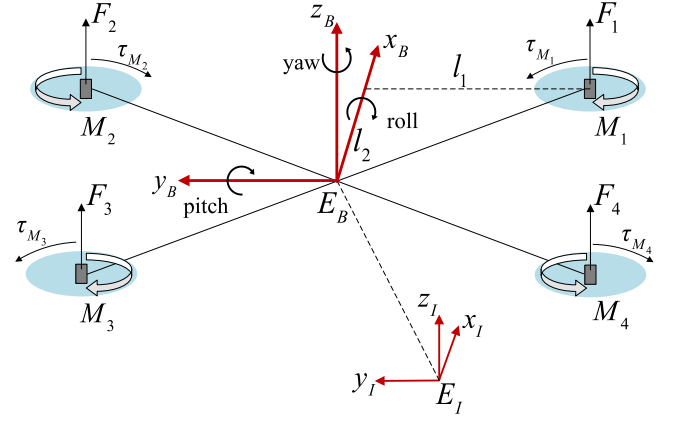


Fig. 1. Quadrotor control model.

that has four control input channels and six degrees of freedom. It is mainly composed by an airframe attached with four motor-driven propellers [26]. Different acts of motions are accomplished by the change of rotational speed of propellers, which produces thrust and torque. The attitude model of quadrotor is introduced in details.

The airframe is regarded as a rigid body, and the geometric center is assumed to be consensus with the center of gravity for sake of convenience. A simplified control model is shown in Fig. 1.

According to Fig. 1, the cross configuration of control mode is utilized. As described in [27], the rotation of four propellers M_i ($i = 1, \dots, 4$) generates thrusts F_i ($i = 1, \dots, 4$) along z -direction of body frame. The relationship between angular speed of propellers Ω_i ($i = 1, \dots, 4$) and F_i is written as $F_i = k_T \Omega_i^2$, where $k_T > 0$ denotes the coefficient related with aerodynamics. The reaction torques τ_{M_i} ($i = 1, \dots, 4$) are generated on airframe by each rotor $\tau_{M_i} = k_D \Omega_i^2$, where $k_D > 0$ denotes the coefficient similarly with k_T .

The roll torque τ_ϕ is produced by the difference of forces $F_2 + F_3$ and $F_1 + F_4$, the pitch torque τ_θ is produced by the difference of forces $F_3 + F_4$ and $F_1 + F_2$, and the yaw torque τ_ψ is produced by the differences of reaction torques τ_{M_i} . The whole relationship between angular speed of propellers and input torques is summarized as follows:

$$\tau = \begin{bmatrix} \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} = \begin{bmatrix} -l_1 k_T & l_1 k_T & l_1 k_T & -l_1 k_T \\ -l_2 k_T & -l_2 k_T & l_2 k_T & l_2 k_T \\ k_D & -k_D & k_D & -k_D \end{bmatrix} \begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix}$$

where l_1 and l_2 are the distances from one propeller to the x_B and y_B axis, respectively.

For convenience of modeling and analyzing, two coordinate frames are introduced: the inertial frame $E_I : \{x_I, y_I, z_I\}$ and the body-fixed frame $E_B : \{x_B, y_B, z_B\}$. The orientation of quadrotor with respect to the inertial frame E_I is described by Euler angles $\Theta = [\phi \ \theta \ \psi]^T$, where ϕ , θ , ψ are the roll, pitch, yaw angle around the x -axis, y -axis, and z -axis, respectively. The angular velocities expressed in the body-fixed frame E_B are written as $\omega = [\omega_x \ \omega_y \ \omega_z]^T$, respectively. According to [28], the

kinematic relationship between ω and $\dot{\Theta}$ is shown as follows:

$$\omega = W_{\Theta} \dot{\Theta} \quad (1)$$

where

$$W_{\Theta} = \begin{bmatrix} 1 & 0 & -\sin \theta \\ 0 & \cos \phi & \cos \theta \sin \phi \\ 0 & -\sin \phi & \cos \theta \cos \phi \end{bmatrix}.$$

Then, the following attitude kinematic model of quadrotor can be derived from (1)

$$\dot{\phi}(t) = \omega_x(t) + \tan \theta \sin \phi \cdot \omega_y(t) + \tan \theta \cos \phi \cdot \omega_z(t) \quad (2)$$

$$\dot{\theta}(t) = \cos \phi \cdot \omega_y(t) - \sin \phi \cdot \omega_z(t) \quad (3)$$

$$\dot{\psi}(t) = \frac{\sin \phi}{\cos \theta} \cdot \omega_y(t) + \frac{\cos \phi}{\cos \theta} \cdot \omega_z(t). \quad (4)$$

Here, for avoiding singularity, θ is assumed to satisfy that $-\pi/2 < \theta < \pi/2$.

Assume that the mass distribution of quadrotor is symmetrical, which makes the rotational inertia matrix $I = \text{diag}\{I_x, I_y, I_z\}$ diagonal. The unmodeled dynamics including parameter uncertainty, gyroscopic effect, and disturbances are expressed in $d = [d_x \ d_y \ d_z]^T$. Then, the attitude dynamic model of quadrotor is derived as follows:

$$\dot{\omega}_x(t) = \frac{\tau_{\phi}(t)}{I_x} + \omega_y(t)\omega_z(t)\frac{I_y - I_z}{I_x} + d_x(t) \quad (5)$$

$$\dot{\omega}_y(t) = \frac{\tau_{\theta}(t)}{I_y} + \omega_x(t)\omega_z(t)\frac{I_z - I_x}{I_y} + d_y(t) \quad (6)$$

$$\dot{\omega}_z(t) = \frac{\tau_{\psi}(t)}{I_z} + \omega_x(t)\omega_y(t)\frac{I_x - I_y}{I_z} + d_z(t). \quad (7)$$

The quadrotor attitude model is divided into three channels for convenience of analyzing, which can be called ϕ - ω_x control channel, θ - ω_y control channel, and ψ - ω_z control channel, respectively. Each control channel of the quadrotor system can be designed similarly and separately.

Remark 1: It is worth mentioning that in actual stable flight control of quadrotor, the variation of angle speed is small, which leads to the coupled terms of the control channels small enough in most circumstance to be estimated and compensated by ESO. Based on the abovementioned observations, the control channels are designed separately in this article.

III. CONTROL ARCHITECTURE

In this section, an inner–outer loop composite disturbance rejection control strategy for attitude control is proposed. The control method on ϕ - ω_x control channel is taken as an example throughout this article. The controller design for other two channels can be performed in a similar way. Note that (5) can be rewritten as a first-order system

$$\dot{x}_1(t) = bu_1(t) + f(t) \quad (8)$$

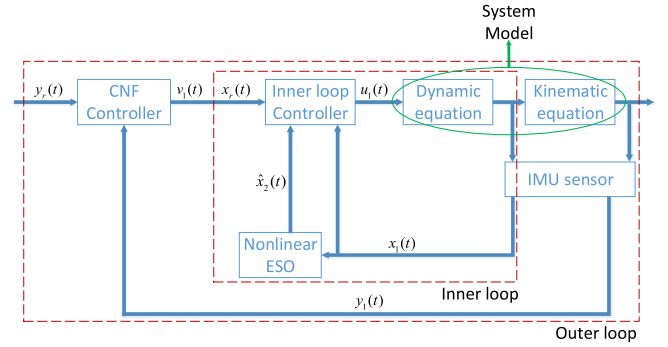


Fig. 2. Diagram of inner–outer loop for attitude control of quadrotor.

where $x_1(t) = \omega_x(t)$, $b = \frac{1}{I_x}$, $u_1(t) = \tau_{\phi}(t)$, and $f(t) = \omega_y(t)\omega_z(t)\frac{I_y - I_z}{I_x} + d_x(t)$. Similarly, (2) can be rewritten as follows:

$$\dot{y}_1(t) = cy_1(t) + f_0(t) \quad (9)$$

where $y_1(t) = \phi(t)$, $c > 0$ denotes an adjustable coefficient, $f_0(t) = \omega_x(t) + \tan \theta \sin \phi \cdot \omega_y(t) + \tan \theta \cos \phi \cdot \omega_z(t) - cy_1(t)$ denotes the nonlinear part of the angular model. The output of outer-loop controller $v_1(t)$ is used as the reference input of inner loop $x_r(t)$. The whole control diagram is presented in Fig. 2.

In general, the inner-loop controller is designed to actively reject disturbance and the outer-loop controller is designed to improve the total tracking performance of the quadrotor attitude. The inertial measurement unit (IMU) is used to measure the angular velocity $x_1(t)$ and angle $y_1(t)$ of the quadrotor. The NESO in Fig. 2 is used to estimate the lumped nonlinearity and disturbance $f(t)$, its output $\hat{x}_2(t)$ is transmitted to the inner-loop controller. The inner-loop controller includes a linear feedback term, a disturbance rejection term and a robust estimation compensation term. The CNF controller includes a linear feedback term and a nonlinear term, which can further improve the angular tracking performance.

Remark 2: Note that, in actual implementation of quadrotor attitude control, the angular velocities ω_x , ω_y , and ω_z are bounded, and the pitch angle θ satisfies $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. Moreover, the inner-loop output $u_1(t)$ and the outer-loop output $v_1(t)$ are also limited in the range of actuator. Therefore, we can assume that $f_0(t)$ is bounded, i.e., there exists $\bar{f}_0$ satisfying $|f_0(t)| \leq \bar{f}_0$ according to the definition of $f_0(t)$, which would be used in outer-loop stability analysis.

It is clear that the stability of both inner- and outer-loop systems does not implies the stability of the whole systems. According to Fig. 2, the output $v_1(t)$ of the outer-loop system can be viewed as the reference input of the inner-loop system. Then, we can design the inner and outer-loop controllers such that the convergence speed of the inner-loop system is much faster than the outer-loop system. According to the time-scale separation hypothesis, the stability of outer loop is approximately equivalent to the stability of the whole system. Based on the abovementioned analysis, the stability of inner loop and outer loop will be analyzed in the following two sections.

IV. INNER-LOOP CONTROL

A. Design of NESO

According to [29], the following assumption is given first in this section.

Assumption 1: The lumped nonlinearity and disturbance $f(t)$ is differentiable and its derivative is bounded, which means that $f(t)$ satisfies $|\dot{f}(t)| \leq \delta$, where δ is a positive constant.

Setting the $f(t)$ as an extended state $x_2(t)$ and $\dot{f}(t) = h(t)$. Then, the dynamic model of roll angle is rewritten as follows:

$$\dot{x}_1(t) = x_2(t) + bu_1(t) \quad (10)$$

$$\dot{x}_2(t) = h(t). \quad (11)$$

For the extended system (10) and (11), the following NESO is designed to estimate the lumped nonlinearity and disturbance

$$\dot{\hat{x}}_1(t) = \hat{x}_2(t) + \frac{l_1}{\epsilon}(x_1(t) - \hat{x}_1(t)) + bu_1(t) \quad (12)$$

$$\dot{\hat{x}}_2(t) = \frac{l_2}{\epsilon^2}(x_1(t) - \hat{x}_1(t)) + \frac{1}{\epsilon}\varphi\left(\omega \cdot \frac{x_1(t) - \hat{x}_1(t)}{\epsilon}\right) \quad (13)$$

where $\hat{x}_1(t)$, $\hat{x}_2(t)$ are the estimates of $x_1(t)$ and $x_2(t)$, respectively, l_1 , l_2 , ϵ , and ω are positive adjustable observer gains. $\varphi(\chi)$ is a nonlinear function satisfying

$$\varphi(\chi) = \begin{cases} -1, & \chi \leq -\frac{\pi}{2} \\ \sin(\chi), & -\frac{\pi}{2} \leq \chi \leq \frac{\pi}{2} \\ 1, & \chi \geq \frac{\pi}{2} \end{cases}.$$

Let the estimation error be $\tilde{x}_1(t) = x_1(t) - \hat{x}_1(t)$ and $\tilde{x}_2(t) = x_2(t) - \hat{x}_2(t)$. According to the abovementioned definitions, the error system is described as follows:

$$\begin{aligned} \dot{\tilde{x}}_1(t) &= \tilde{x}_2(t) - l_1 \frac{\tilde{x}_1(t)}{\epsilon} \\ &= \tilde{x}_2(t) - g_1\left(\frac{\tilde{x}_1(t)}{\epsilon}\right) \end{aligned} \quad (14)$$

$$\begin{aligned} \dot{\tilde{x}}_2(t) &= -\frac{l_2}{\epsilon} \cdot \frac{\tilde{x}_1(t)}{\epsilon} - \frac{1}{\epsilon}\varphi\left(\omega \cdot \frac{\tilde{x}_1(t)}{\epsilon}\right) + h(t) \\ &= -\frac{1}{\epsilon}g_2\left(\frac{\tilde{x}_1(t)}{\epsilon}\right) + h(t) \end{aligned} \quad (15)$$

where $g_1(\chi)$ and $g_2(\chi)$ are functions as

$$g_1(\chi) = l_1\chi, \quad g_2(\chi) = l_2\chi + \varphi(\omega\chi).$$

By setting $\eta_1(t) = \frac{\tilde{x}_1(\epsilon t)}{\epsilon}$, $\eta_2(t) = \tilde{x}_2(\epsilon t)$, the expression of $\eta(t) = [\eta_1(t), \eta_2(t)]^T$ is given as

$$\dot{\eta}_1(t) = \eta_2(t) - g_1(\eta_1(t)) \quad (16)$$

$$\dot{\eta}_2(t) = -g_2(\eta_1(t)) + \epsilon h(\epsilon t). \quad (17)$$

By choosing $W(y) = \lambda_3 \|y\|^2$ in [29, Assumption (H3)], the stability conditions of the NESO tracking error system (14) and (15) are listed in the following lemma.

Lemma 1: If there exist positive constants λ_i ($i = 1, 2, 3$), β , and a positive function $V \in \mathbb{R}$ satisfying

$$\lambda_1 \|\eta\|^2 \leq V(\eta) \leq \lambda_2 \|\eta\|^2 \quad (18)$$

$$\frac{\partial V}{\partial \eta_1}(\eta_2(t) - g_1(\eta_1(t))) - \frac{\partial V}{\partial \eta_2}g_2(\eta_1(t)) \leq -\lambda_3 \|\eta\|^2 \quad (19)$$

$$\left| \frac{\partial V}{\partial \eta_2} \right| \leq \beta \|\eta\| \quad (20)$$

then for every positive constant $N > 0$, we have

$$\lim_{\epsilon \rightarrow 0} |\tilde{x}_i(t)| \rightarrow 0, \quad \text{uniformly in } t, t > N$$

$$\text{and } \lim_{t \rightarrow \infty} |\tilde{x}_i(t)| \leq O(\epsilon^{3-i}), \quad i = 1, 2.$$

According to (16) and (17), it is obtained that

$$\dot{\eta}(t) = A\eta(t) + \zeta(t) \quad (21)$$

where

$$A = \begin{bmatrix} -l_1 & 1 \\ -l_2 & 0 \end{bmatrix}, \zeta(t) = \begin{bmatrix} 0 \\ \epsilon h(\epsilon t) - \varphi(\omega \eta_1(t)) \end{bmatrix}.$$

Note that A is Hurwitz for the given $l_1 > 0$ and $l_2 > 0$, thus there exists a positive definite matrix P being the solution of the Lyapunov function $PA + A^T P = -I_2$, and P can be determined as

$$P = \begin{bmatrix} \frac{1+l_2}{2l_1} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{l_1^2+l_2+1}{2l_1 l_2} \end{bmatrix}.$$

Consider the Lyapunov function as

$$V(\eta(t)) = \frac{l_1 l_2}{l_1^2 + l_2 + 1} \eta^T(t) P \eta(t) + \int_0^{\eta_1(t)} \varphi(s) ds.$$

We can easily get

$$\frac{\partial V}{\partial \eta_1(t)} = \frac{l_1 l_2}{l_1^2 + l_2 + 1} \left[\frac{1+l_2}{l_1} \eta_1(t) - \eta_2(t) \right] + \varphi(\eta_1(t))$$

and

$$\frac{\partial V}{\partial \eta_2(t)} = \eta_2(t) - \frac{l_1 l_2}{l_1^2 + l_2 + 1} \eta_1(t).$$

Then, we have

$$\begin{aligned} & \frac{\partial V}{\partial \eta_1}(\eta_2(t) - g_1(\eta_1(t))) - \frac{\partial V}{\partial \eta_2}g_2(\eta_1(t)) \\ &= -\frac{l_1 l_2}{l_1^2 + l_2 + 1} (\eta_1^2(t) + \eta_2^2(t)) - \frac{l_1^3 + l_1}{l_1^2 + l_2 + 1} \eta_1(t) \varphi(\eta_1(t)) \\ &\leq -\frac{l_1 l_2}{l_1^2 + l_2 + 1} (\eta_1^2(t) + \eta_2^2(t)) \\ &= -\lambda_3 \|\eta\|^2. \end{aligned}$$

On the other hand

$$\begin{aligned} \left| \frac{\partial V}{\partial \eta_2} \right| &= \left| -\frac{l_1 l_2}{l_1^2 + l_2 + 1} \eta_1(t) + \eta_2(t) \right| \\ &\leq \sqrt{1 + \left(\frac{l_1 l_2}{l_1^2 + l_2 + 1} \right)^2} \|\eta(t)\| \\ &= \beta \|\eta(t)\|. \end{aligned}$$

Thus, all conditions of Lemma 1 are satisfied without any other limitations. Therefore, the proposed NESO with suitable parameters can estimate the system state and the lumped nonlinearity and disturbance effectively.

Remark 3: In order to design the controller, rotational inertia of quadrotor is needed for calculating $b = 1/I_x$. If the information is not precisely known, the estimate of $b_0 = 1/I_{x0}$ can be used, and the term $(b - b_0)u(t)$ should be merged into $f(t)$.

B. Design of Inner-Loop Disturbance Rejection Controller

The disturbance rejection controller includes two parts: linear feedback term and estimation error compensation term.

First, a linear feedback controller including disturbance rejection term is designed as

$$u_L(t) = -k(x_1(t) - x_r(t)) - \frac{1}{b}\hat{x}_2(t) \quad (22)$$

where $k > 0$ is the state feedback gain and $x_r(t)$ is the reference input. In the inner loop, the reference input $x_r(t)$ can be viewed as the output of CNF controller $v_1(t)$.

From the proof of [29, Th. 2.1], we can get

$$\begin{aligned} |\tilde{x}_2(t)| &\leq \frac{\sqrt{\lambda_1 V(\eta(0))}}{\lambda_1} e^{-\frac{\lambda_3 t}{2\lambda_2 \epsilon}} + \frac{\epsilon \beta \delta}{2\lambda_1} \int_0^t e^{-\frac{\lambda_3}{2\lambda_2}(\frac{t}{\epsilon} - s)} ds \\ &\leq \frac{\sqrt{\lambda_1 V(\eta(0))}}{\lambda_1} + \frac{\epsilon \beta \delta \lambda_2}{\lambda_1 \lambda_3}. \end{aligned}$$

Without loss of generality, the upper bound of $\dot{x}_r(t)$ is assumed to be known as $|\dot{x}_r(t)| \leq \sigma_r$. Then, the following equation is derived

$$|\tilde{x}_2(t)| + |\dot{x}_r(t)| \leq \frac{\sqrt{\lambda_1 V(\eta(0))}}{\lambda_1} + \frac{\epsilon \beta \delta \lambda_2}{\lambda_1 \lambda_3} + \sigma_r \triangleq \sigma$$

where $\sigma > 0$ is a known constant. A robust compensation term is incorporated to improve the disturbance rejection performance [30]

$$u_D(t) = -\varphi(\omega_0 e(t)) \cdot \frac{\sigma}{b} \quad (23)$$

where $e(t)$ is the output tracking error defined by $e(t) = x_1(t) - x_r(t)$.

Finally, by combining (22) and (23), we get the composite inner-loop controller as follows:

$$u_1(t) = u_L(t) + u_D(t). \quad (24)$$

C. Stability Analysis of the Inner-Loop System

In the following theorem, the stability of the inner-loop system is analyzed. Substituting controller (24) into system (8) yields the following closed-loop system

$$\begin{aligned} \dot{x}_1(t) &= b[u_L(t) + u_D(t)] + f(t) \\ &= -bk(x_1(t) - x_r(t)) - \hat{x}_2(t) + f(t) \\ &\quad - \sigma \varphi(\omega_0 \cdot e(t)). \end{aligned} \quad (25)$$

Theorem 1: Consider system (25) and the NESO (12) and (13), for given controller parameters $k > 0$ and $\omega_0 > 0$, the output tracking error $e(t)$ is UUB.

Proof: In the case of $|e(t)| \geq \frac{\pi}{2\omega_0}$, we can get the following output tracking error equation from (25)

$$\dot{e}(t) = -bke(t) + \tilde{x}_2(t) - \dot{x}_r(t) - \sigma \text{sign}(e(t)).$$

Consider the Lyapunov function $V(t) = \frac{1}{2}e^2(t)$, the derivative function can be calculated as

$$\begin{aligned} \dot{V}(t) &= e(t)\dot{e}(t) \\ &= -bke^2(t) + \tilde{x}_2(t)e(t) - \dot{x}_r(t)e(t) - \sigma|e(t)| \\ &\leq -bke^2(t) + (|\tilde{x}_2(t)| + |\dot{x}_r(t)| - \sigma)|e(t)| \\ &< 0. \end{aligned}$$

Thus, the output tracking error $e(t)$ will tend to zero asymptotically until the condition $|e(t)| \geq \frac{\pi}{2\omega_0}$ is not satisfied.

In the case of $|e(t)| < \frac{\pi}{2\omega_0}$, we can get the output tracking error equation from (25)

$$\dot{e}(t) = -bke(t) + \tilde{x}_2(t) - \dot{x}_r(t) - \sigma \sin(\omega_0 e(t)).$$

Then,

$$\begin{aligned} \dot{V}(t) &= e(t)\dot{e}(t) \\ &= -bke^2(t) + \tilde{x}_2(t)e(t) - \dot{x}_r(t)e(t) \\ &\quad - \sigma e(t) \sin(\omega_0 e(t)) \\ &\leq -bke^2(t) + (|\tilde{x}_2(t)| + |\dot{x}_r(t)|) \cdot |e(t)| \\ &\leq -bke^2(t) + \sigma|e(t)|. \end{aligned}$$

We can conclude that if $|e(t)| \geq \frac{\sigma}{bk}$, $\dot{V}(t) \leq 0$. Thus, $|e(t)|$ is bounded with the equation

$$|e(t)| \leq \min \left(\frac{\sigma}{bk}, \frac{\pi}{2\omega_0} \right) \quad (26)$$

which is related with the parameter k and ω_0 .

Remark 4: It is easily concluded from (26) that the bound of tracking error would reduce when k and ω_0 is increased. However, the overlarge k would result in system oscillation, and the function $\varphi(\cdot)$ would tend to $\text{sign}(\cdot)$ with $\omega_0 \rightarrow \infty$, which leads to the discontinuity of u_D . Thus, the parameter k and ω_0 should be chosen in a suitable range.

V. OUTER-LOOP CONTROL

The objective of the outer-loop control system is to calculate an appropriate target angular velocity, which makes the angle of quadrotor tends to the desired angle.

A. Design of Outer-Loop CNF Controller

The composite nonlinear feedback (CNF) controller is designed to improve the angular tracking performance [31]. It includes two parts: linear feedback term and nonlinear feedback term. First, a linear feedback controller is designed

$$v_L(t) = -k_0 (y_1(t) - y_r(t)) + \frac{\dot{y}_r(t)}{c} \quad (27)$$

where $k_0 > 0$ is the state feedback gain and $y_r(t)$ is the reference input. $y_r(t)$ can be designed to get a desired attitude.

Next, to improve the tracking performance, we add a nonlinear feedback term

$$v_N(t) = \rho(y_r(t), y_1(t)) \cdot (y_1(t) - y_r(t)) \quad (28)$$

where $\rho(y_r(t), y_1(t))$ is a nonpositive function of $y_1(t)$ and $y_r(t)$, which is locally Lipschitz, and changes with the output feedback error $y_1(t) - y_r(t)$. When system output approaches the reference input, $\rho(y_r(t), y_1(t))$ changes to improve the transient performance of the system. A normally used form is designed as [32]

$$\rho(y_r(t), y_1(t)) = -\beta e^{-\alpha \alpha_0(t) |y_1(t) - y_r(t)|} \quad (29)$$

where

$$\alpha_0(t) = \begin{cases} \frac{1}{|y_0 - y_r(t)|}, & y_0 \neq y_r(t) \\ 1, & y_0 = y_r(t) \end{cases}$$

and y_0 is the initial value of $y_1(t)$, $\alpha > 0$, and $\beta > 0$ are two adjustable coefficients.

Remark 5: From the nonlinear term (29), we can conclude that if $y_0 \neq y_r(t)$, the function $\rho(y_r(t), y_1(t))$ changes from $-\beta e^{-\alpha}$ to $-\beta$ while $y_1(t)$ approaches $y_r(t)$. With the appropriate parameters α and β , the nonlinear part accounts for a small proportion of the total control at the initial condition, and the linear control term plays a major role at that time. When system state approaches to the desired value, the nonlinear control term action gradually increases to speed up the regulation process.

By combining (27) and (28), the CNF controller is written as follows:

$$v_1(t) = v_L(t) + v_N(t). \quad (30)$$

B. Stability Analysis of the Outer-Loop System

With the stability of the inner-loop system in hand, $x_1(t)$ in (9) can be replaced by $v_1(t)$. Substituting the controller (30) into the roll angle system model (9), we can get

$$\begin{aligned} \dot{y}_1(t) &= c(v_L(t) + v_N(t)) + f_0(t) \\ &= -ck_0(y_1(t) - y_r(t)) + \dot{y}_r(t) \\ &\quad + c\rho(y_r(t), y_1(t)) \cdot (y_1(t) - y_r(t)) + f_0(t). \end{aligned} \quad (31)$$

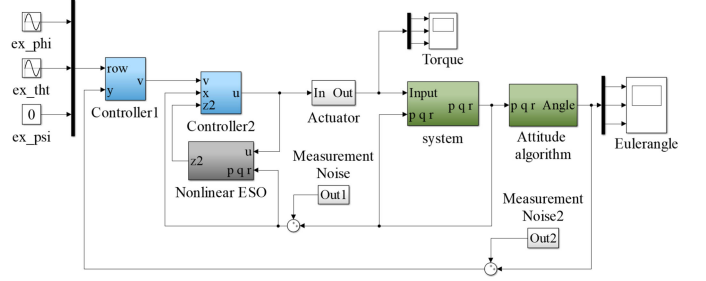


Fig. 3. Simulink control structure.

Define the output tracking error $\varepsilon(t)$ as $\varepsilon(t) = y_1(t) - y_r(t)$. Then, we have the following theorem.

Theorem 2: Consider system (31). For a given controller parameters $c > 0$ and $k_0 > 0$, the output tracking error is UUB.

Proof: It is easy to obtain that

$$\dot{\varepsilon}(t) = -ck_0\varepsilon(t) + c\varepsilon(t)\rho(y_r(t), y_1(t)) + f_0(t).$$

According to Remark 2, the nonlinear function $f_0(t)$ is bounded with $|f_0(t)| \leq \bar{f}_0$. Consider the Lyapunov function $W(t) = \frac{1}{2}\varepsilon^2(t)$, then its derivative is calculated as

$$\begin{aligned} \dot{W}(t) &= \varepsilon(t)\dot{\varepsilon}(t) \\ &= (-ck_0 + c\rho(y_r(t), y_1(t)))\varepsilon^2(t) + f_0(t)\varepsilon(t). \end{aligned}$$

Since $\rho(y_r(t), y_1(t)) < 0$, we have

$$\begin{aligned} \dot{W}(t) &= (-ck_0 + c\rho(y_r(t), y_1(t)))\varepsilon^2(t) + f_0(t)\varepsilon(t) \\ &\leq -ck_0\varepsilon^2(t) + \bar{f}_0|\varepsilon(t)|. \end{aligned}$$

Then, we can get easily that, when $|\varepsilon(t)|$ is out of the range of $(0, \frac{\bar{f}_0}{ck_0})$, $\dot{W}(t) \leq 0$ always holds. Thus, $|\varepsilon(t)|$ is bounded with the equation

$$|\varepsilon(t)| \leq \frac{\bar{f}_0}{ck_0}$$

which is related with the parameter c and k_0 .

VI. SIMULATION AND EXPERIMENTAL RESULTS

A. Simulation Results

In this section, a numerical simulation is performed in MATLAB 2016a to show the effectiveness of the proposed control method. The data of actual quadrotor is used. The rotational inertia in three axis is $I_x = 2.7 \times 10^{-3} \text{ kg}\cdot\text{m}^2$, $I_y = 2.6 \times 10^{-3} \text{ kg}\cdot\text{m}^2$, and $I_z = 4.7 \times 10^{-3} \text{ kg}\cdot\text{m}^2$.

A Simulink model is built, as shown in Fig. 3. The initial conditions are $[\phi_0 \ \theta_0 \ \psi_0] = [0^\circ \ -12^\circ \ 0^\circ]$ and $[\omega_{x0} \ \omega_{y0} \ \omega_{z0}] = [0 \ 0 \ 0]$. All initial states of NESO are set to be zero. The reference signals represented by angles are provided as follows:

$$\phi_r(t) = 12 \cdot \sin\left(\frac{\pi}{10}t\right), \ \theta_r(t) = 12 \cdot \sin\left(\frac{\pi}{10}t - \frac{\pi}{2}\right), \ \psi_r(t) = 0.$$

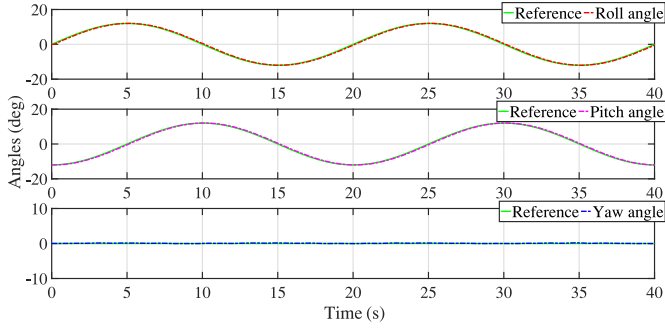


Fig. 4. Angle tracking result.

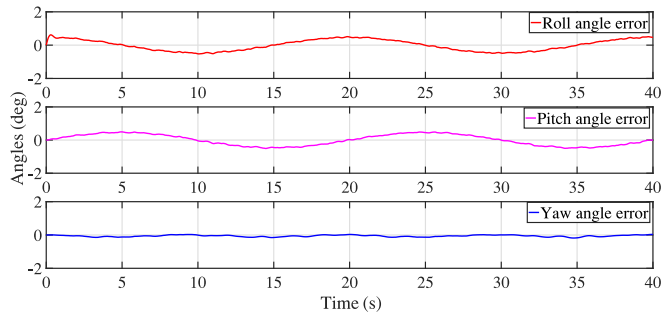


Fig. 5. Angle tracking error.

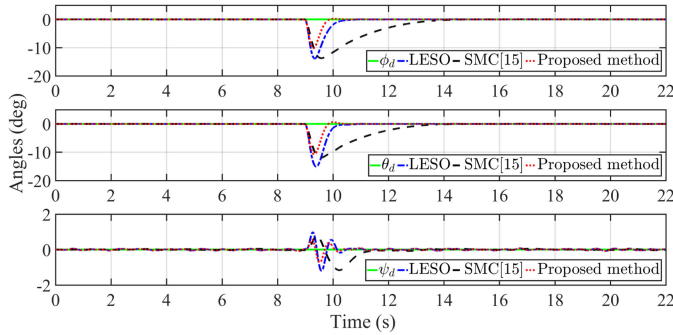


Fig. 6. Comparison of the angle stabilization performance.

In practical test, there is always a measurement noise from the data of sensor. Thus, a white noise with zero mean and covariance of 0.5° is added in angle feedback loop, and a white noise with zero mean and standard deviation of 0.1° is added in angular velocity feedback loop, respectively. The simulation results are shown in Figs. 4 to 5.

It is concluded from Fig. 4 that the proposed method has a better precision of attitude tracking despite of noises.

Another simulation is implemented to show the disturbance rejection performance. Two step signals of -0.06 and $-0.05 \text{ N} \cdot \text{m}$ are added on the roll torque and pitch torque channels based on the abovementioned simulation. Under the same conditions, the comparative simulation results among the LESO based approach, SMC approach, and the proposed control approach in this article are shown in Fig. 6, which further shows the better disturbance rejection performance of the proposed method.



Fig. 7. Attitude control platform.

B. Experimental Results

The quadrotor used in experiment is composed by QAV-250 frame, ANO-Pioneer flight control board and actuator. The flight control board carries the STM32F407VG microcontroller and ICM20602 IMU module, which can process the signal of gyroscope and accelerometer, calculate the output torque and send them to the actuator in one millisecond. The actuator includes EMAX MT2204 KV2300 DC motors, EMAX Sinonk 12A electronic speed controllers, and 5045 self-tightening propellers. A 2200 mAh 3 s 45C LIPO battery is used to provide electricity. The software used for programming is Keil MDK version 5. The ANO TC ground station is used to transport data between quadrotor and computer. The control cycles of inner loop and outer loop are 1 ms and 10 ms, respectively.

Based on the abovementioned quadrotor, an attitude control platform, as shown in Fig. 7, is built to verify the effectiveness of the proposed method. The quadrotor is fixed on a cardan joint to restrict its translational movement. Two kinds of verification experiments are given as follows.

1) *Attitude Tracking Performance*: In this part, a set of experiments are designed to evaluate the tracking performance of the proposed method in comparison of the traditional control method. The reference signals, parameters of controller and NESO are in consensus with simulation part. The experimental results are shown in Figs. 8–11.

According to Fig. 8, it is clear that the actual angles of quadrotor can track the reference signals well. Fig. 9 shows that the tracking errors have been successfully restricted near the origin besides the small oscillation caused by measurement noise, sampling frequency, and airflow. The variables x_1 and z_1 are shown in Fig. 11. It indicates that the proposed NESO can estimate the system state well. The comparative simulation results shown in Fig. 8 have further verified the efficiency of the proposed method.

2) *Disturbance Rejection Performance*: It is shown in first part that the proposed NESO can track the system state

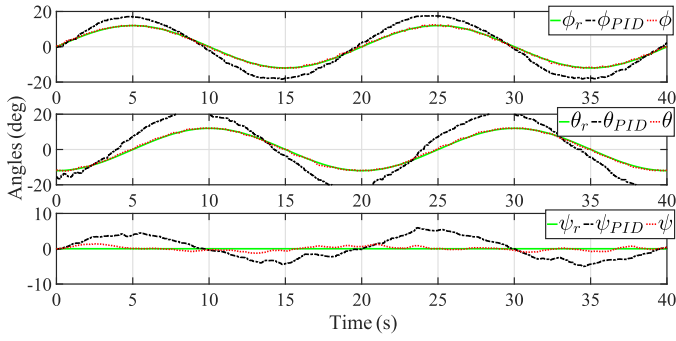


Fig. 8. Attitude tracking performance based on the proposed method and double-loop PID method.

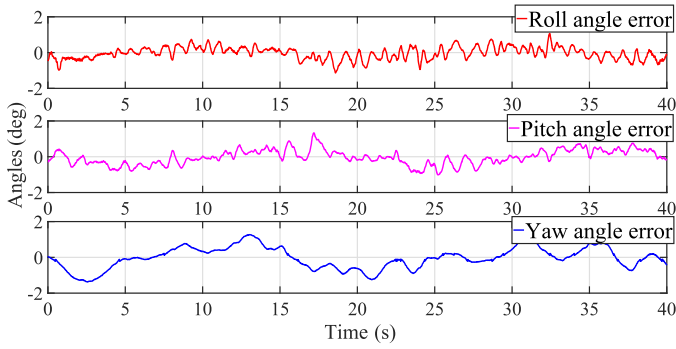


Fig. 9. Tracking error.

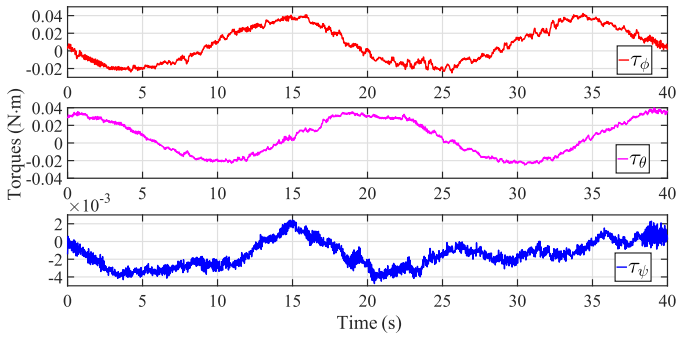


Fig. 10. Controller output.

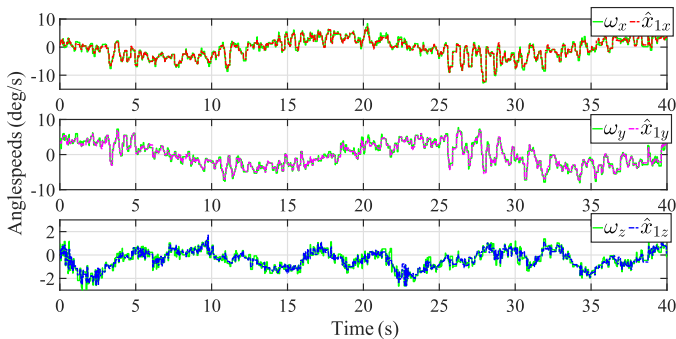


Fig. 11. ESO state tracking performance.

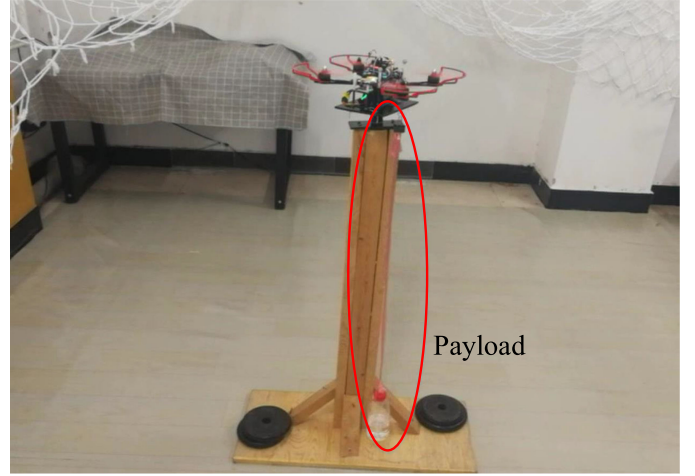


Fig. 12. Attitude control platform with payload.

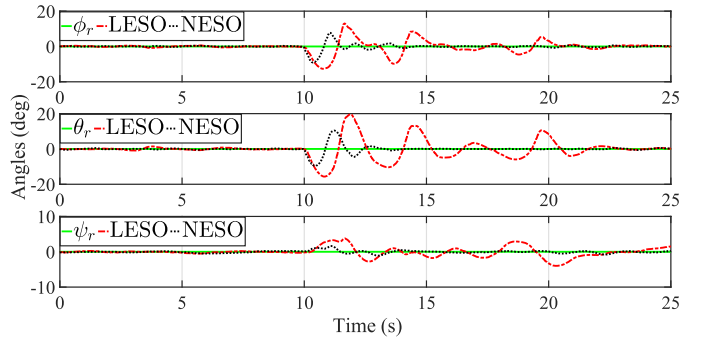


Fig. 13. Attitude tracking performance with disturbance.

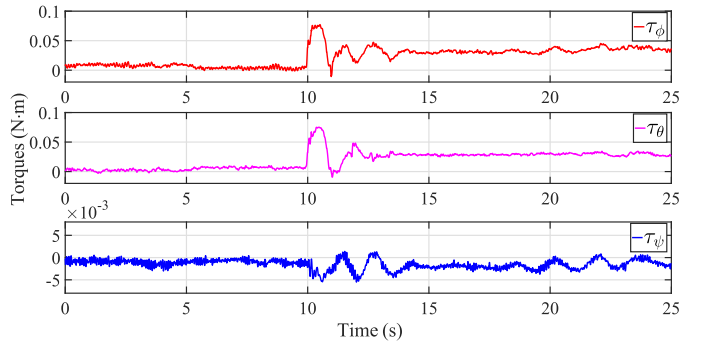


Fig. 14. Controller output with disturbance.

effectively. To show its efficiency in disturbance estimation and compensation, another experiment is designed in this part. A payload (a bottle of water) is used to generate a step disturbance torque. It is taken in one side of quadrotor at 9 s when the quadrotor is flying with a stable attitude. The experiment is shown in Fig. 12. All parameters are inherited from the previous experiments. The results are shown in Figs. 13 and 14. If we set the nonlinear term $\varphi(\cdot) = 0$ in (13), the NESO (13) can be reduced into a traditional LESO, and the comparative results between the NESO and LESO under the same method are given

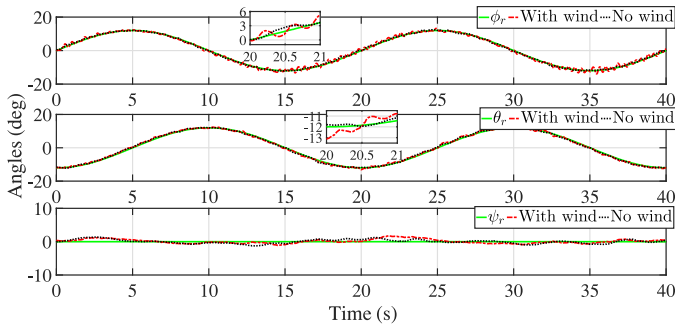


Fig. 15. Attitude tracking performance with gust wind.

in Fig. 13, which indicates that the quadrotor recovers quickly from the disturbance torque. It is also clear in Fig. 14 that the controller has a quick response and precise compensation acting on control torques.

3) *Overall Performance*: In order to simulate the true flight conditions, an electric fan is put aside of the quadrotor platform to generate a gust wind. The comparative results for the attitude tracking with and without gust wind are shown in Fig. 15, which further indicates the strong robustness of the proposed method under variant conditions.

It follows from the abovementioned experiments that the proposed control strategy can achieve the attitude stabilization, angle tracking, and disturbance rejection effectively.

VII. CONCLUSION

In this article, attitude control problem of quadrotor was investigated. Based on the attitude dynamic model with disturbance, a NESO-based disturbance rejection control strategy was developed, and a double-loop composite control architecture was established. The inner loop and CNF controllers can not only attenuate disturbance but also improve the angular tracking performance of quadrotor. Simulations and experimental results proved the efficiency of the proposed method. In this article, the basic attitude stabilization problem was considered for quadrotors, and some interesting problems, such as trajectory tracking control, and cooperative control of multiquadrotors may be addressed in the future works.

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